

BBSO Invitational

Heredity

Division B

Key

Genetic Code

		Second Letter					
		U	C	A	G		
First Letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA } Stop UAG }	UGU } Cys UGC } UGA } Stop UGG } Trp	U	C
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U	C
	A	AUU } AUC } Ile AUA } AUG } Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U	C
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } Asp GGC } GGA } Glu GGG }	U	C
		Third Letter					

Use the following gene information to answer questions

For Gene A, A - 2 eyes and a - 3 eyes

For Gene B, B - 2 arms and b - 3 arms

For Gene C, C - 2 legs and c - 3 legs

2. (3) You cross two individuals with the genotypes Aabb x AABb. What are the genotypic and phenotypic ratios? (Order from largest to smallest numerically. Ex: 3:2:1 not 1:2:3 or 2:1:3)

+1.5 Genotypic: 1:1:1:1

+1.5 Phenotypic: 1:1

3. (1 each) Predict the probability of the following genotypes and phenotypes of the offspring from aaBbCCDD x AaBbCcdd.

(a) AABBCC 0

(b) AaBBCC 1/16

(c) aaBbCC 1/8

(d) 2 eyes, 2 arms, 2 legs 3/8

(e) 3 arms 1/4

(f) 3 eyes, 3 arms, 3 legs 0

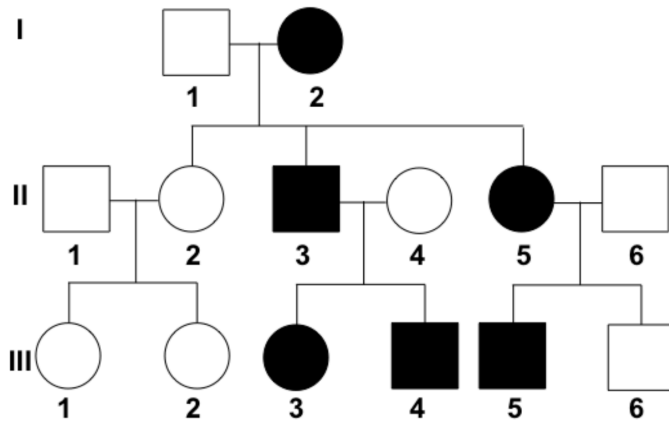
4. (2) You cross two individuals that are both heterozygous for genes A through Z, which is 26 genes in total. What is the probability that an offspring is also heterozygous for genes A-Z? Give your answer as an expression. (Don't calculate anything).

+2 $(\frac{1}{2})^{26}$

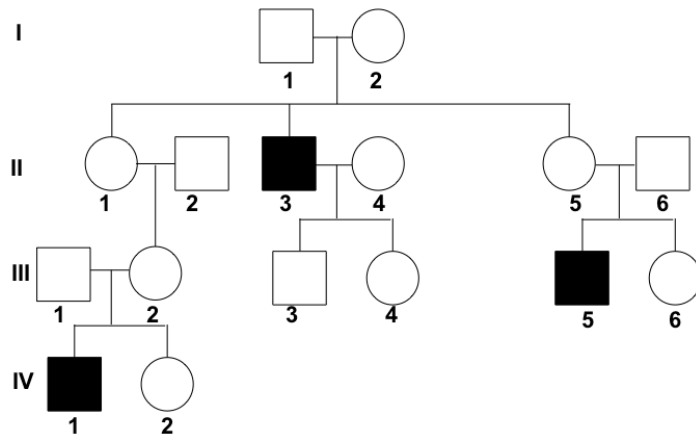
5. (2) A male with blood type AB is crossed with a female of the genotype $I^B i$. What is the probability that the offspring is type B?

+2 1/2

8. (2 each) What modes of inheritance are displayed in the pedigrees?



a. +2 autosomal dominant



b. +2 X-linked recessive

-1.5 each if answer includes extra words that change meaning of answer (ex: autosomal dominant/recessive would receive only .5 pts for the left one)

Mitosis and Meiosis

9. (1) What is the order of the stages in mitosis?

+1 Prophase, (prometaphase), metaphase, anaphase, telophase

10. (1) What are the stages of the cell cycle in order?

+1 G1, S, G2, M

11. (2) A cell ($2n = 8$) undergoes meiosis 1 and meiosis 2. How many chromosomes do the daughter cells have after meiosis 2?

+2 4

12. (2) What phenomenon is responsible for the above daughter cells being genetically different?

+2 Crossing over

13. (1 each) Give the structures to the following descriptions in mitosis and meiosis

- a. DNA sequence that holds sister chromatids together _____ **centromere** _____
- b. Attaches chromosomes to microtubules _____ **kinetochore** _____
- c. Pulls chromosomes apart in anaphase _____ **spindle fibers**
(microtubules/cytoskeleton) _____
- d. Microtubule-organizing centers _____ **centrosome** _____
- e. Equivalent of cleavage furrow in plant cells _____ **cell plate** _____

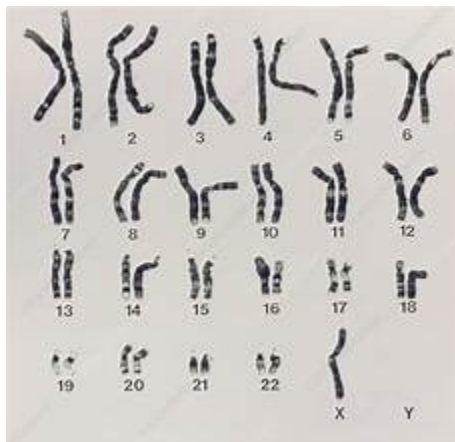
14. (2) What is the term for uneven splitting of chromosomes during anaphase?

+ 2 nondisjunction

15. (1) Which of Mendel's laws does nondisjunction violate?

+1 Law of segregation

16. (1) What is the genetic disease depicted in the karyotype below?

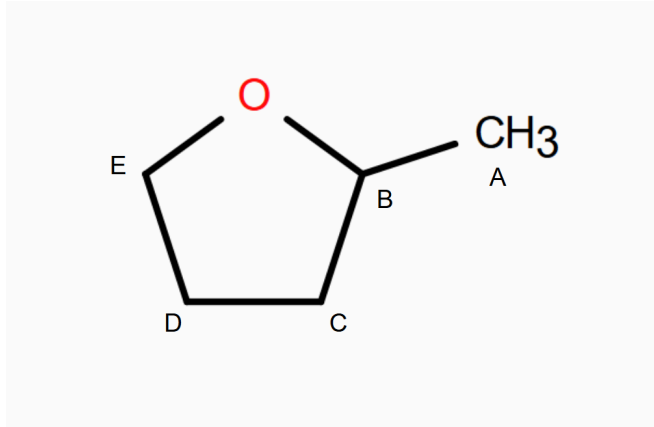


+ 1 Turner syndrome

17. (1) What are the implications of an absent Y chromosome in the individual?

+1 The individual is biologically female. Anything on the lines of adopting feminine features or absence of male features is also ok.

Molecular Biology of DNA



18. (3) The pentose sugar of a nucleotide is shown. What three structures are attached to the pentose sugar which would fully comprise a nucleotide? (You can ignore the letters)

+1 each Phosphate, nitrogenous base, hydroxide

19. (1) How are deoxyriboses in DNA different from riboses in RNA?

+1 RNA has extra OH

20. (2) Given the DNA sequence 5'- GATACAGATACA - 3'. What is the complementary DNA strand?

+2 5' - TGTATCTGTATC - 3'

21. (1 each) Match the enzyme with its function in DNA replication.

1. Helicase
2. Single-stranded binding proteins
3. Primase
4. DNA Polymerase
5. Ligase
6. Topoisomerase

- a. 2 Binds to separated single DNA strands
- b. 4 Adds nucleotides to synthesize new DNA strands
- c. 5 Joins Okazaki fragments by forming phosphodiester bonds

- d. 1 Unwinds DNA double helix
- e. 3 Synthesizes RNA primers for initiation of DNA synthesis
- f. 6 Relieves tension generated from DNA supercoiling

22. (3) Rank the following in order from least compact to most compact. Just use the numbers.

- 1. Euchromatin
- 2. Chromosomes
- 3. DNA double helix
- 4. Heterochromatin

+3 3, 1, 4, 2

Prokaryotic and eukaryotic gene expression and regulation

Fill in the following blanks

24. (4) If a DNA double helix has 32% adenine, what percent of thymine, guanine, cytosine, and uracil does it contain?

+1 each 32% thymine 18% cytosine 18% guanine 0% uracil

25. (5) The enzyme responsible for transcription is RNA polymerase, and this process takes place in the nucleus. The 3 stages of transcription are initiation, elongation, termination.

26. (5) Translation occurs in the cytoplasm (location) where it is carried out by ribosomes (organelle). In the A site, the tRNA enters, and the polypeptide is transferred from the tRNA already in the P site. Both tRNAs shift one site over, and the tRNA that previously had the polypeptide exits through the E site.

27. (1) Name an enzyme that violates the Central Dogma.

+1 Reverse transcriptase or RNA-dependent DNA polymerase or telomerase

28. (2) What is the name of the phenomenon that allows for multiple codons to code for the same amino acid?

+2 Wobble

30. (3) Given the template strand below, what would be the resulting amino acid sequence?

TACGCATAACTGGTGTACATT

- a. (1) What is the mRNA transcribed from the template strand

+1 AUGCGUAUUGACCACAUGUAA

- b. (2) What is the amino acid sequence translated from the mRNA

+2 Met, Arg, Ile, Asp, His, Met

Technology and techniques

31. (3) What are the 3 stages of PCR, and what temperatures are they performed at?

+5 Denaturation +5 94 C, +5 Annealing +5 50-65 C, +5 Extension +5 72 C
+- 3 for temps still good for full credit

32. (2) What special DNA polymerase is used in PCR?

+2 taq

33. (1) DNA polymerase can not initiate DNA synthesis by itself. What does polymerase add onto to synthesize DNA in PCR?

+1 primers

34. (2) How many DNA double helices are present after one DNA molecule undergoes 5 PCR rounds?

+2 32

35. (1) What did one DNA double helix say to the other DNA double helix? (this is meant to be a joke)

+1 anything is fine

36. (1) What are you interested in competing in Heredity B? (wrong answers only)

+1 anything is fine